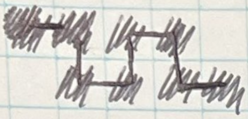
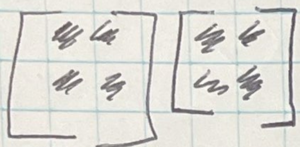




```
moves = [ ]
coords = [ ]

for seg in segments:
    for block in range(len(seg)):
        if len(coords) == 0
            cords.append((0,0))
            moves.append((1,0))      (# first moves doesn't matter)
        else:
            last_coord = coord[-1]
            last move = moves[-1]
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

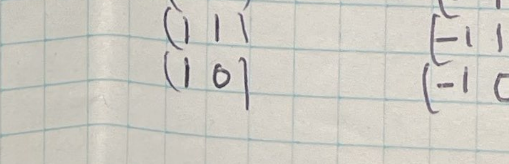
segments = [2, 2]

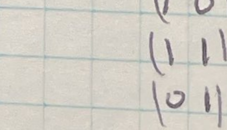
$X_0 = (0,0)$

$X_1 = (0,1)$ or $(1,0)$

$X_2 =$

| | |
| --- | --- |
| $X_2$: $(1,1)$ or $(-1,1)$ | $X_2$: $(1,1)$ or $(1,-1)$ |
| $X_3 = (1,2)$ $(-1,2)$ | $(2,1)$ $(2,-1)$ |
| $(1,0)$ $(-1,0)$ | $(0,1)$ $(0,-1)$ |



$\begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{pmatrix}$ $\begin{pmatrix} 0 & 0 \\ 0 & 1 \\ -1 & 1 \\ -1 & 0 \end{pmatrix}$ $\begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}$ $\begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 1 & -1 \\ 0 & -1 \end{pmatrix}$

moves = []
coords = []

for seg in segments:
    i = 0
    ~~for block in rang([illegible](seg)):~~
    while i < seg:
        if len(coords) == 0:

            coords.append((0,0))
            moves.append((1,0))
            continue
        [illegible]:
        last_coord = cords[-1]

        last_move = moves[-1]

    [illegible]

        new_move = rotate(last_move)

        new_coord = last_coord + new_move
        Check if new coord is allowed
        moves.append(new_move)
        coords.append(new_coord)


def new_coord_is_allowed(new_coord, [illegible] coords [illegible], max_dim=2):

    if new_coord in coords:
        return False

    if sum(abs(x_coords)) > max_dim or
    sum(abs(y_coords)) > max_dim:
        return False
return true